

## 多元随机变量及其分布

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1. (1) 由题可知:

$$\begin{cases} P(XY \neq 0) = P(X \neq 0, Y \neq 0) = a + c + 0.2 = 0.4 \\ P(X \leq 0 | Y \leq 0) = \frac{P(X \leq 0, Y \leq 0)}{P(Y \leq 0)} = \frac{a+b+0.1}{a+b+0.3} = \frac{2}{3} \\ a + b + c = 0.4 \end{cases}$$

解得  $a = 0.1, b = 0.2, c = 0.1$ .

(2)  $P(X = -1) = 0.3, P(X = 0) = 0.4, P(X = 1) = 0.3$ .

$P(Y = -1) = 0.2, P(Y = 0) = 0.4, P(Y = 1) = 0.4$ .

(3)  $X + Y \in \{-2, -1, 0, 1, 2\}$ .

$P(X + Y = -2) = 0.1, P(X + Y = -1) = 0.1, P(X + Y = 0) = 0.4, P(X + Y = 1) = 0.3, P(X + Y = 2) = 0.1$ .

2. (1) 根据概率密度的定义可知:

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$$

故

$$\int_{-1}^1 \int_{x^2}^1 cx^2 y dx dy = 1$$

解得  $c = \frac{21}{4}$ .

(2) 根据边缘概率密度的定义知:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy, f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

故

$$f_X(x) = \frac{21}{4} \int_{x^2}^1 x^2 y dy = \frac{21}{8} x^2 (1 - x^4), x \in [-1, 1]$$

$$f_Y(y) = \frac{21}{4} \int_{-\sqrt{y}}^{\sqrt{y}} x^2 y dx = \frac{7}{2} y^{\frac{5}{2}}, y \in [0, 1]$$

综上

$$f_X(x) = \begin{cases} \frac{21}{8}x^2(1-x^4), & x \in [-1, 1] \\ 0, & \text{else} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{7}{2}y^{\frac{5}{2}}, & y \in [0, 1] \\ 0, & \text{else} \end{cases}$$

3. 由题可知:

$$f_{X_1, X_2, X_3|X}(1, 0, 0 | 1) + f_{X_1, X_2, X_3|X}(0, 1, 0 | 1) + f_{X_1, X_2, X_3|X}(0, 0, 1 | 1) = p_1 + p_2 + p_3 = 1$$

且

$$f_X(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

故

$$\begin{aligned} f_{X_1, X_2, X_3}(x_1, x_2, x_3) &= e^{-\lambda} \lambda^x \frac{p_1^{x_1} p_2^{x_2} p_3^{x_3}}{x_1! x_2! x_3!} \\ &= e^{-\lambda} \frac{(p_1 \lambda)^{x_1}}{x_1!} \cdot \frac{(p_2 \lambda)^{x_2}}{x_2!} \cdot \frac{(p_3 \lambda)^{x_3}}{x_3!} \\ &= \left[ e^{-p_1 \lambda} \frac{(p_1 \lambda)^{x_1}}{x_1!} \right] \cdot \left[ e^{-p_2 \lambda} \frac{(p_2 \lambda)^{x_2}}{x_2!} \right] \cdot \left[ e^{-p_3 \lambda} \frac{(p_3 \lambda)^{x_3}}{x_3!} \right] \\ &= f(x_1) f(x_2) f(x_3) \end{aligned}$$

显然有  $f_{X_i}(x_i) = f(x_i), i = 1, 2, 3$ . 故  $X_1, X_2, X_3$  相互独立且  $X_i \sim \pi(p_i \lambda), i = 1, 2, 3$ .  $\square$

4. (1) 设随机变量  $Z_1 = Xf(Y)$ , 则

$$Z_1 = \begin{cases} X, & Y \in A \\ -X, & Y \notin A \end{cases}$$

故  $Z_1$  的分布函数为

$$\begin{aligned} F_{Z_1}(z_1) &= P(Z_1 \leq z_1) \\ &= P(X \leq z_1 | Y \in A) + P(-X \leq z_1 | Y \notin A) \\ &= P(X \leq z_1 | Y \in A) + P(X \leq z_1 | Y \notin A) \\ &= P(X \leq z_1) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_1} e^{-\frac{x^2}{2}} dx \\ &= \Phi(z_1) \end{aligned}$$

(2) 设随机变量  $Z_2 = f(Y)$ , 则

$$\begin{aligned} F_{Z_1, Z_2}(z_1, z_2) &= P(Z_1 \leq z_1, Z_2 \leq z_2) \\ &= \begin{cases} 0, & z_2 < -1 \\ P(Z_1 \leq z_1, Z_2 = -1), & -1 \leq z_2 < 1 \\ P(Z_1 \leq z_1), & z_2 \geq 1 \end{cases} \end{aligned}$$

另一方面,

$$F_{Z_2}(z_2) = \begin{cases} 0, & z_2 < -1 \\ P(Z_2 = -1), & -1 \leq z_2 < 1 \\ 1, & z_2 \geq 1 \end{cases}$$

故只需证

$$P(Z_1 \leq z_1, Z_2 = -1) = P(Z_1 \leq z_1)P(Z_2 = -1)$$

由第 (1) 小问及  $X, Y$  相互独立知:

$$\begin{aligned} P(Z_1 \leq z_1, Z_2 = -1) &= P(X \leq z_1, Y \notin A) \\ &= P(X \leq z_1)P(Y \notin A) \\ &= P(Z_1 \leq z_1)P(Z_2 = -1) \end{aligned}$$

综上,

$$F_{Z_1, Z_2}(z_1, z_2) = F_{Z_1}(z_1)F_{Z_2}(z_2)$$

$Z_1, Z_2$  相互独立. □

5. (1) 根据概率分布函数的定义:

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= \int_{x+y \leq z} f(x, y) dx dy \\ &= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{z-x} f(x, y) dy \end{aligned}$$

则

$$\begin{aligned} f_Z(z) &= \frac{dF_Z(z)}{dz} = \int_{-\infty}^{+\infty} f(x, z-x) dx \\ &= \begin{cases} \int_{\max\{0, z-1\}}^{\min\{1, z\}} z dx, & 0 < x < 1, 0 < z-x < 1 \\ 0, & \text{else} \end{cases} \\ &= \begin{cases} z^2, & 0 < z \leq 1 \\ z(2-z), & 1 < z < 2 \\ 0, & \text{else} \end{cases} \end{aligned}$$

(2) 与第 (1) 小问同理可得:

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{+\infty} f(x, \frac{z}{x}) \cdot \frac{1}{x} dx \\ &= \begin{cases} \int_z^1 (1 + \frac{z}{x^2}) dx, & 0 < x < 1, 0 < \frac{z}{x} < 1 \\ 0, & \text{else} \end{cases} \\ &= \begin{cases} 2-2z, & 0 < z < 1 \\ 0, & \text{else} \end{cases} \end{aligned}$$

6. (1) 由题可知:

$$\begin{aligned}
 P(X = k_1, Z = k_2) &= \begin{cases} 0, & k_1 > k_2 \\ P(X = k_1, Y \leq k_1), & k_1 = k_2 \\ P(X = k_1, Y = k_2), & k_1 < k_2 \end{cases} \\
 &= \begin{cases} 0, & k_1 > k_2 \\ (1-p)^{k_1-1}p(1-(1-p)^{k_1}), & k_1 = k_2 \\ p^2(1-p)^{k_1+k_2-2}, & k_1 < k_2 \end{cases}
 \end{aligned}$$

(2) 由题可知:

$$\begin{aligned}
 P(Z = k) &= P(X = k, Y \leq k) + P(Y = k, X < k) \\
 &= (1-p)^{k-1}p(1-(1-p)^k) + (1-p)^{k-1}p(1-(1-p)^{k-1}) \\
 &= (1-p)^{k-1}p \left[ 2 - (1-p)^k - (1-p)^{k-1} \right]
 \end{aligned}$$

则关于  $Z$  的条件概率分布为

$$\begin{aligned}
 P(X = k_1 | Z = k_2) &= \frac{P(X = k_1, Z = k_2)}{P(Z = k_2)} \\
 &= \begin{cases} 0, & k_1 > k_2 \\ \frac{1-(1-p)^{k_1}}{2-(1-p)^{k_1}-(1-p)^{k_1-1}}, & k_1 = k_2 \\ \frac{p(1-p)^{k_1-1}}{2-(1-p)^{k_2}-(1-p)^{k_2-1}}, & k_1 < k_2 \end{cases}
 \end{aligned}$$

7. 考虑映射  $h: (R, \Theta, \Phi) \mapsto (X_1, X_2, X_3)$ , 其中  $R \in [0, +\infty), \Theta \in [0, \pi], \Phi \in [0, 2\pi]$ ,

$$X_1 = R \sin \Theta \sin \Phi$$

$$X_2 = R \sin \Theta \cos \Phi$$

$$X_3 = R \cos \Theta$$

则有  $R = \sqrt{X_1^2 + X_2^2 + X_3^2}$ .

由概率密度变换公式:

$$\begin{aligned}
 f_{R,\Theta,\Phi}(r, \theta, \phi) &= f_{X_1, X_2, X_3}(x_1, x_2, x_3) |\det \mathbf{J}(r, \theta, \phi)| \\
 &= \frac{1}{(\sqrt{2\pi}\sigma)^3} e^{-\frac{x_1^2 + x_2^2 + x_3^2}{2\sigma^2}} (r^2 \sin \theta) \\
 &= \frac{r^2 \sin \theta}{(\sqrt{2\pi}\sigma)^3} e^{-\frac{r^2}{2\sigma^2}}
 \end{aligned}$$

故

$$\begin{aligned}
 f_R(r) &= \left( \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \right) \frac{r^2}{(\sqrt{2\pi}\sigma)^3} e^{-\frac{r^2}{2\sigma^2}} \\
 &= \frac{2r^2}{\sqrt{2\pi}\sigma^3} e^{-\frac{r^2}{2\sigma^2}}, \quad r \geq 0
 \end{aligned}$$

综上,

$$f_R(r) = \begin{cases} \frac{2r^2}{\sqrt{2\pi}\sigma^3} e^{-\frac{r^2}{2\sigma^2}}, & r \geq 0 \\ 0, & \text{else} \end{cases}$$

8. (1) 考虑映射  $h: (z, w) \mapsto (x, y)$ , 其中  $z \in (0, +\infty), w \in (\sqrt{z}, +\infty)$ ,

$$x = \sqrt{z}$$

$$y = w - \sqrt{z}$$

由概率密度变换公式:

$$\begin{aligned} f_{Z,W}(z, w) &= f_{X,Y}(x, y) |\det \mathbf{J}(z, w)| \\ &= \lambda^2 e^{-\frac{x+y}{\lambda}} \frac{1}{2} z^{-\frac{1}{2}} \\ &= \frac{\lambda^2}{2\sqrt{z}} e^{-\lambda w}, z \in (0, +\infty), w \in (\sqrt{z}, +\infty) \end{aligned}$$

综上,

$$f_{Z,W}(z, w) = \begin{cases} \frac{\lambda^2}{2\sqrt{z}} e^{-\lambda w}, & z \in (0, +\infty), w \in (\sqrt{z}, +\infty) \\ 0, & \text{else} \end{cases}$$

- (2) 由条件概率密度的定义:

$$f_{Z|W}(z | w) = \frac{f_{Z,W}(z, w)}{f_W(w)}$$

由边缘密度的定义:

$$\begin{aligned} f_W(w) &= \int_0^{w^2} f_{Z,W}(z, w) dz \\ &= \lambda^2 w e^{-\lambda w}, \quad w \in (0, +\infty) \end{aligned}$$

故

$$f_{Z|W}(z | w) = \begin{cases} \frac{1}{2w\sqrt{z}}, & z \in (0, +\infty), w \in (\sqrt{z}, +\infty) \\ 0, & \text{else} \end{cases}$$

9. (1) 考虑映射  $h_1: (y_1, y_2) \mapsto (x_1, x_2)$ ,

$$x_1 = \frac{y_1}{\cos \theta} x_2 = \frac{y_2}{\sin \theta}$$

根据随机变量的独立性及概率密度变换公式可得:

$$f_{Y_1, Y_2}(y_1, y_2) = f\left(\frac{y_1}{\cos \theta}\right) f\left(\frac{y_2}{\sin \theta}\right) \left| \frac{1}{\cos \theta \sin \theta} \right|$$

故

$$f_{Z_1}(z_1) = \int_{-\infty}^{+\infty} f\left(\frac{y_1}{\cos \theta}\right) f\left(\frac{z_1 - y_1}{\sin \theta}\right) \left| \frac{1}{\cos \theta \sin \theta} \right| dy_1$$

(2) 考虑映射  $h_2 : (x_1, x_2, x_3) \mapsto (z_1, z_2, z_3)$ ,

$$z_1 = x_1 \cos \theta + x_2 \sin \theta$$

$$z_2 = x_1 \sin \theta + x_3 \cos \theta$$

$$z_3 = x_3$$

则有

$$x_1 = \frac{z_2 - z_3 \cos \theta}{\sin \theta}$$

$$x_2 = \frac{z_1}{\sin \theta} - \frac{z_2 \cos \theta - z_3 \cos^2 \theta}{\sin^2 \theta}$$

$$x_3 = z_3$$

根据随机变量的独立性及概率密度变换公式可得：

$$\begin{aligned} f_{Z_1, Z_2, Z_3}(z_1, z_2, z_3) &= f_{X_1, X_2, X_3}(x_1, x_2, x_3) \left| \frac{1}{\sin^2 \theta} \right| \\ &= f\left(\frac{z_2 - z_3 \cos \theta}{\sin \theta}\right) f\left(\frac{z_1}{\sin \theta} - \frac{z_2 \cos \theta - z_3 \cos^2 \theta}{\sin^2 \theta}\right) f(z_3) \left| \frac{1}{\sin^2 \theta} \right| \end{aligned}$$

故

$$f_{Z_1, Z_2}(z_1, z_2) = \int_{-\infty}^{+\infty} f\left(\frac{z_2 - z_3 \cos \theta}{\sin \theta}\right) f\left(\frac{z_1}{\sin \theta} - \frac{z_2 \cos \theta - z_3 \cos^2 \theta}{\sin^2 \theta}\right) f(z_3) \left| \frac{1}{\sin^2 \theta} \right| dz_3$$